IN SWACHH BHARAT ENVIRONMENTAL GAMIFICATION

Clean & Earn

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Transforming individual cleanliness efforts into verified, rewarded, and habitual civic action using mobile image capture, GPS geotagging, and Google Gemini Multimodal Vision AI.

📦 Download Source Code (.zip)

📄 Download PDF + Word (.docx)

🌐 Live Web App ↗

Core Stack: Next.js 16 • Tailwind CSS • Supabase PostgreSQL • Google Gemini Vision AI • Vercel

SLIDE 2 — MOTIVATION & ORIGIN

Why Clean & Earn?

"Cleanliness is a shared responsibility, yet people do not always participate actively in keeping their surroundings clean."

THE CORE CATALYST QUESTION

"How can we motivate people to take responsibility for cleanliness and make it a regular habit?"

1. PROBLEM OBSERVED

Limited motivation for consistent community participation and unrewarded civic efforts.

2. THE SOLUTION IDEA

Make cleanliness engaging through structured missions, photo evidence, and a tangible Green Points economy.

3. THE CIVIC PURPOSE

Encourage citizens to turn small individual cleanup actions into lasting, shared community habits.

CORE GUIDING THOUGHT: What if doing something good for the environment also became rewarding, verifiable, and habitual?

SLIDE 3 — PROBLEM & PROPOSED SOLUTION

Turning an Idea into Action

⚠️ CURRENT PROBLEMS IDENTIFIED

- Awareness exists, but consistent motivation to participate is lacking.
- Individual cleanup efforts often go unrecognised, unrecorded, and uncelebrated.
- Claims of completed activities lack an objective, fraud-proof verification mechanism.

✨ CLEAN & EARN SOLUTION

- Mobile web-first platform accessible on any device without mandatory app store downloads.
- Dual-photo proof (Before & After) with automatic hardware GPS lock.
- Google Gemini AI verification verifies authenticity and awards Green Points instantly.

THE CLOSED-LOOP CIVIC ACTION LIFECYCLE

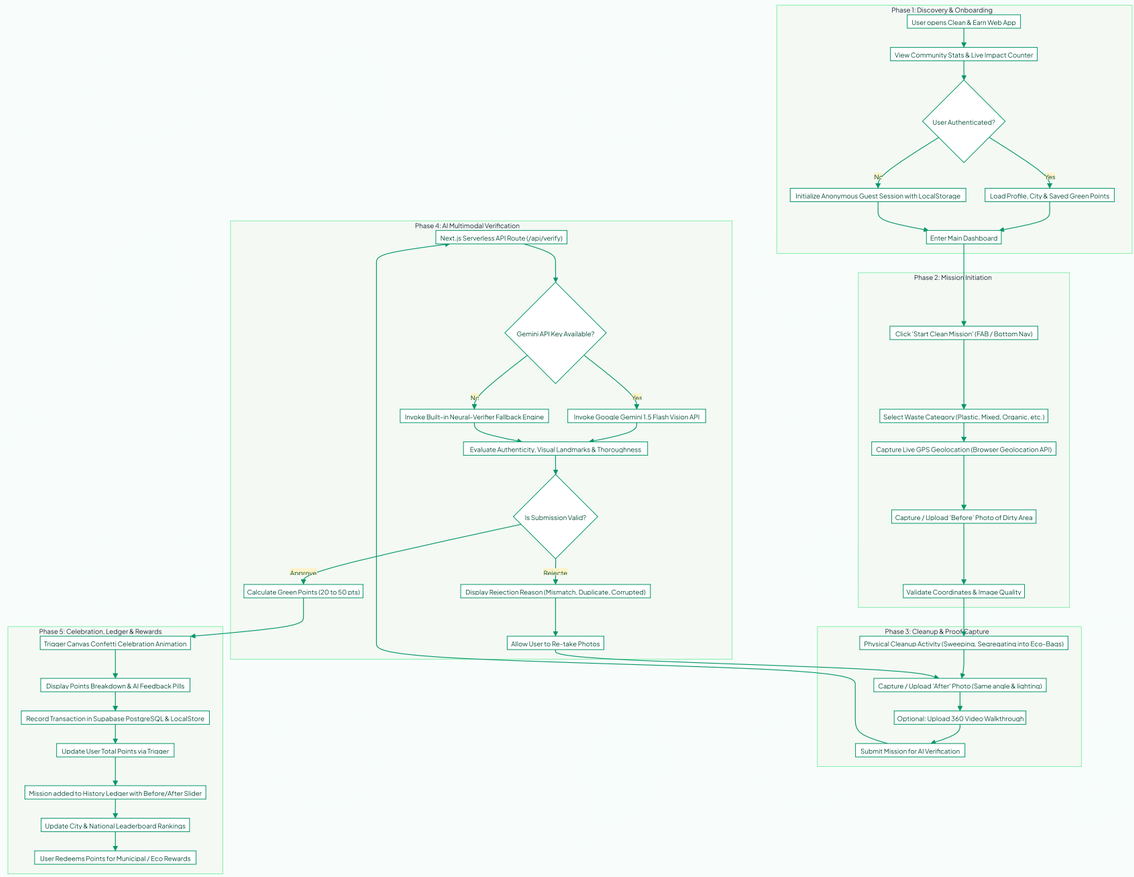
1. Start Mission → 2. Clean Area → 3. Before/After Photos → 4. AI Verification → **5. Green Points!**

Key Architectural Principle: Transform individual cleanliness efforts into documented, auditable, and verifiable community impact.

SLIDE 4 — END-TO-END USER FLOW

Interactive 5-Phase Citizen Journey

`Onboarding → Mission Setup → Physical Action → AI Verification → Rewards`



💡 Drag with mouse/touch • Mouse Wheel: Zoom in/out

1. Discovery
Zero-friction guest session

2. Setup
Category & GPS geotag

3. Cleanup
Physical sweeping & proof

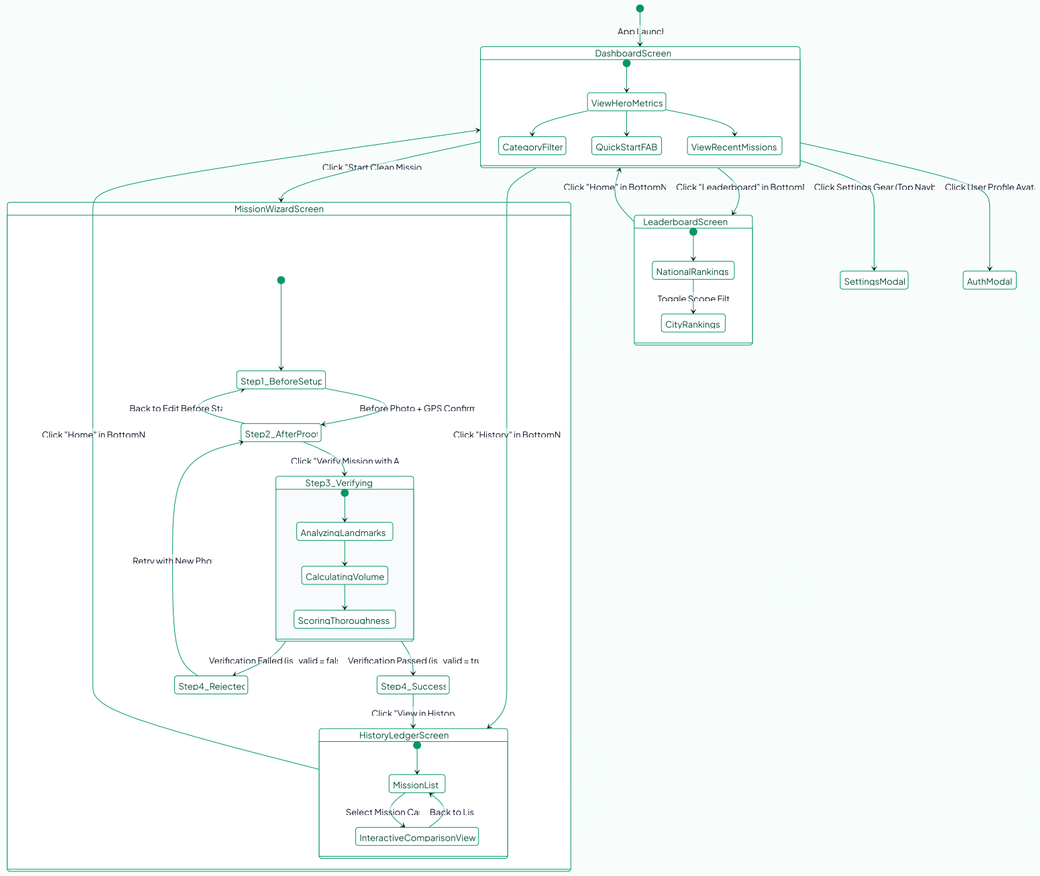
4. Gemini AI
Landmark & volume audit

5. Reward
Ledger & Green Points

SLIDE 5 — UI/UX SCREEN ARCHITECTURE

Mobile Screen Navigation & State Machine

`Dashboard ⇄ MissionWizard (Steps 1-4) ⇄ Ledger ⇄ Leaderboard`



`💡 Drag with mouse/touch • Mouse Wheel: Zoom in/out`

🌞 High Outdoor Legibility

High-contrast sage theme with emerald action pills designed for bright tropical sunlight.

👍 Thumb-Zone First

Primary action buttons and navigation pinned to bottom 30% of screen for one-handed operation.

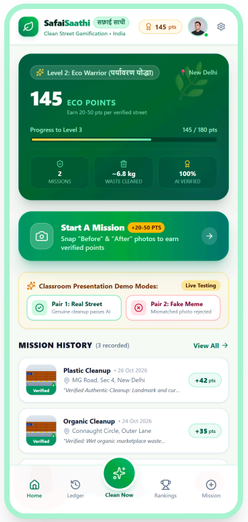
✨ Dopamine Feedback

Dynamic canvas-confetti, live points increment animation, and before/after interactive slider.

SLIDE 6 — PRODUCT WALKTHROUGH

Live Citizen Dashboard

Step 0 — Overview



1. Instant Impact Header

Persistent Green Points balance, Swachh Bharat mascot, and quick city switcher (New Delhi / Mumbai / Bengaluru).

2. Hero Stats & Motivation

Displays total verified missions, estimated kg of waste cleared, and dynamic civic quotes.

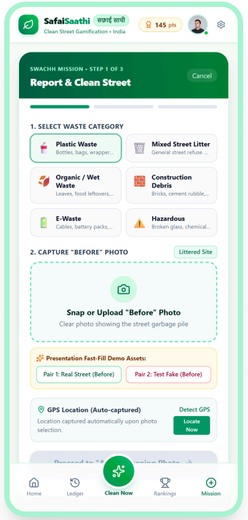
3. 1-Tap Mission Launcher

Prominent "Start Clean Mission" floating action button plus pre-loaded 1-click test demo pairs for testing.

SLIDE 7 — PRODUCT WALKTHROUGH

Mission Step 1 — Categorization & Geotagging

Step 1 – Setup



1. Waste Categorization Grid

Tactile selection for Plastic 🥤, Mixed 🗑️, Organic 🥑, Construction 🧱, E-Waste 🔋, or Hazardous ⚠️.

2. Automatic GPS Geotag Lock

Direct integration with device GPS coordinates (Lat/Long) locks physical location bounds before camera triggers.

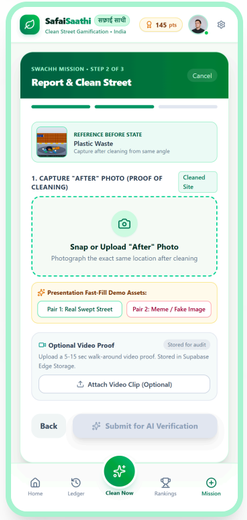
3. "Before" Photo Viewfinder

Direct access to native mobile rear camera with instant client-side thumbnail validation.

SLIDE 8 — PRODUCT WALKTHROUGH

Mission Step 2 — Dual-Proof Capture

`Step 2 — Proof`



1. Side-by-Side Angle Helper

Small floating preview of the "Before" photo assists the user in aligning landmarks (curb lines, walls, cracks).

2. "After" Photo Capture

Captures cleaned street area with bagged waste ready for municipal collection.

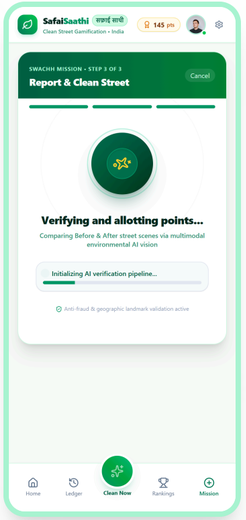
3. Optional Video Audit Field

Supports 360-degree video walkthrough upload for government or corporate CSR audits.

SLIDE 9 — PRODUCT WALKTHROUGH

Mission Step 3 — AI Radar Scanner

`Step 3 — Analysis`



1. Animated Radar Graphic

Pulsing concentric scanner indicates real-time backend vision processing with clear status text.

2. Multi-Stage Visual Verification

"Checking GPS bounds...", "Scanning spatial landmarks...", and "Evaluating trash reduction volume...".

3. Low Cognitive Friction

Verification typically finishes within 1.2 to 2.5 seconds over standard mobile 4G/5G connections.

SLIDE 10 — PRODUCT WALKTHROUGH

Step 4 — Verification Outcomes & Anti-Fraud

Step 4 – Results

OUTCOME A: APPROVED (+44 POINTS)



Canvas-confetti burst, score allotment pill, and detailed vision reasoning breakdown.

OUTCOME B: REJECTED (ANTI-FRAUD)



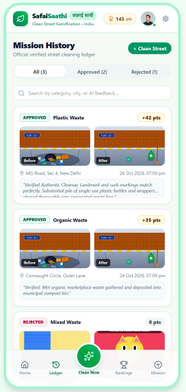
Detects memes, unrelated images, or duplicate before/after files with constructive retry advice.

SLIDE 11 — PRODUCT WALKTHROUGH

Public History Ledger & Civic Leaderboard

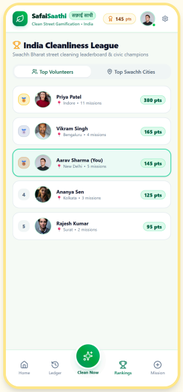
Engagement

INTERACTIVE PROOF LEDGER



Interactive Before/After draggable split slider, timestamps, and Google Maps pin links.

CITY & NATIONAL RANKINGS

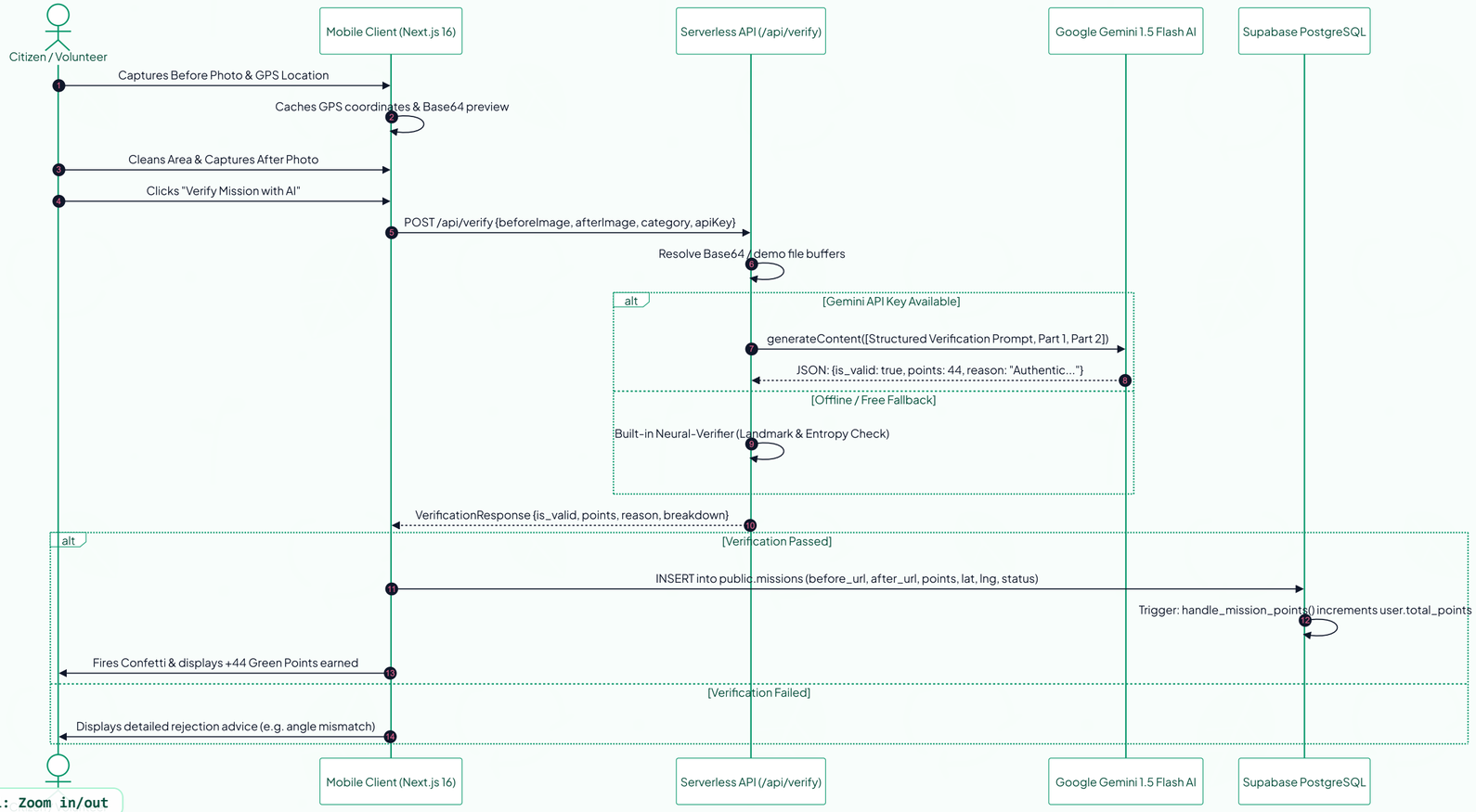


Gold, Silver, Bronze badges, user rank highlight, and city-level cleanliness competitions.

SLIDE 12 — AI PIPELINE & SEQUENCE

Multimodal Vision Verification Sequence

`Client ⇄ Next.js Route ⇄ Gemini 1.5 Vision ⇄ Supabase DB`



`Drag with mouse/touch • Mouse Wheel: Zoom in/out`

Gemini 1.5 Flash Vision

Sub-2s latency multimodal model checks curb lines, background buildings, and sidewalk cracks.

Zero-Cost Fallback Engine

Built-in neural-verifier handles demo pairs and offline evaluation without requiring external keys.

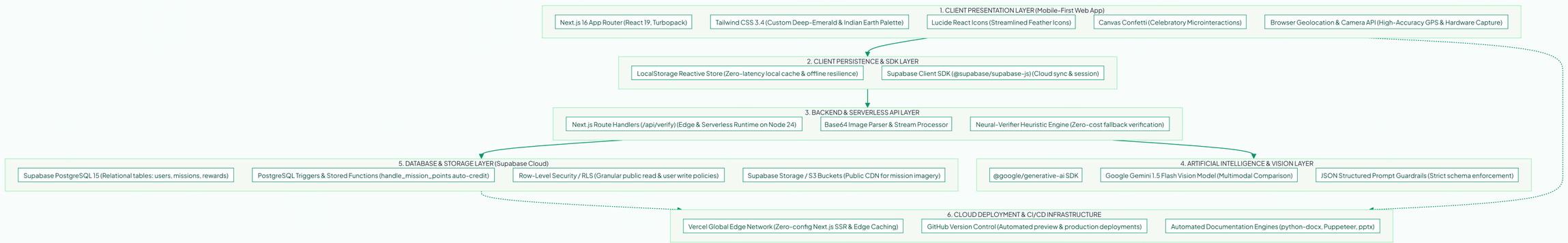
Atomic SQL Trigger

PostgreSQL handle_mission_points() triggers automatically upon approved mission inserts.

SLIDE 13 — SYSTEM ARCHITECTURE

6-Tier System & Technology Stack Architecture

Cloud-Native • Serverless • Edge-Distributed



💡 Drag with mouse/touch • Mouse Wheel: Zoom in/out

1. Presentation	2. Persistence	3. Serverless	4. AI & Vision	5. Database	6. Edge Infra
Next.js 16 & Tailwind	LocalStorage & SDK	Route Handlers (Node 24)	Gemini 1.5 Flash	Supabase SQL & S3	Vercel Global Edge

SLIDE 14 — TECHNOLOGY STACK MATRIX

Full-Stack Engineering Specifications

Download Tech Word (.docx)

TECHNOLOGY	LAYER	ROLE & ARCHITECTURAL CONTRIBUTION
Next.js 16 (React 19, Turbopack)	Full-Stack Framework	App Router architecture, fast server-rendered components, zero-config serverless API routes, and instant static pre-rendering.
TypeScript 5.x	Type Safety	End-to-end type safety guaranteeing schema integrity between database models, Gemini verification payloads, and UI components.
Tailwind CSS 3.4	Design System	Utility-first responsive styles with custom Deep-Emerald and Indian Earth palette, high outdoor sunlight contrast, and dark mode accents.
Google Gemini 1.5 Flash Vision	Multimodal AI	Multimodal vision analysis comparing Before/After photographs, spatial landmark correlation, meme/fraud rejection, and thoroughness scoring.
Supabase PostgreSQL 15	Relational DB	Stores persistent user accounts, mission logs, coordinates, and categories under granular Row-Level Security (RLS) policies.
PostgreSQL Stored Procedures	Database Triggers	Plpgsql triggers (handle_mission_points) execute atomically inside the database to increment user points upon approved missions.
Supabase Storage	Object Storage	S3-compatible public bucket (missions) hosting before and after photo evidence with instant worldwide CDN delivery.
Canvas Confetti	UX Microinteraction	Lightweight canvas-based particle generator triggered immediately upon AI verification approval for maximum psychological reward.

SLIDE 15 — PRACTICAL FEASIBILITY

Can Clean & Earn Work in Real Life?

Yes. The project can begin with a small pilot and expand based on empirical outcomes.

1. TECHNICAL FEASIBILITY

Modern web standards support camera access, GPS tagging, and real-time AI scoring directly in mobile browsers with zero installation friction.

2. OPERATIONAL FEASIBILITY

Initial pilots can run in university campuses or residential housing societies where local supervisors validate eco-disposal points.

3. FINANCIAL FEASIBILITY

Serverless architecture costs $0 at idle. Green Points can initially be backed by campus perks, certificate credits, or corporate CSR grants.

4. SOCIAL FEASIBILITY

Voluntary gamified participation taps into civic pride, community recognition, and shared ownership of public spaces.

FEASIBILITY CONCLUSION: Clean & Earn is ready for a controlled pilot, with wider implementation scaling according to verified community outcomes.

SLIDE 16 — VISION & LONG-TERM ROADMAP

From Project to Community Platform

VISION STATEMENT

"To build a community where cleanliness becomes a shared habit and technology encourages people to create meaningful environmental impact."

Phase 1
Build & Test
Validate core workflows with university pilot users.

Phase 2
Engage
Expand to residential wards & student volunteer clubs.

Phase 3
Partnerships
Partner with NGOs, municipal wards, and reward sponsors.

Phase 4
Improve AI
Fine-tune verification models with empirical field photos.

Phase 5
Scale
Responsible municipal expansion across cities.

Long-Term Goal: Build a sustainable platform that encourages continuous participation rather than one-time cleanups.

SLIDE 17 — FIVE-YEAR IMPACT REVIEW

How Will We Measure Real Success?

1. PARTICIPATION RETENTION

Active & Repeat Volunteers

Measuring weekly repeat cleanups rather than vanity signup numbers. Success means habitual civic action.

2. VERIFIED ACTIVITIES

Audit Precision & Waste Diverted

Number of approved cleanup missions and total metric tons of segregated waste diverted from open streets.

3. OBSERVABLE IMPACT

Cleaner Street Neighborhoods

Longitudinal photographic tracking to verify that cleaned areas remain litter-free over extended periods.

4. LONG-TERM SUSTAINABILITY

Partnerships & Operating Trust

Sustaining reward funds through corporate CSR, municipal grants, and preserving user privacy.

SLIDE 18 — CONCLUSION

Small Actions. Collective Impact.

- Clean & Earn connects community participation with photographic evidence, verification, and incentives.
- Technology provides structure and anti-fraud verification, while human habits create lasting change.
- Next step: Launch an on-ground pilot in a college/community to measure empirical results and refine verification.

"A cleaner environment begins when small individual actions become a shared community habit."

Clean & Earn — Encouraging Action, Rewarding Responsibility.

📦 Download Source Code (.zip)

📄 Download PDF + Word (.docx)

Thank You, Sir! Any Questions? • Live at https://garbage-rosy-six.vercel.app